Version 1.1

Physics PhD Atlas 2027

The International Applicant's Playbook (US, Canada + Europe)

Version 1.1 · program data checked October 21, 2026 · 40 programs (21 US, 10 Canada, 9 Europe: Germany, Italy, Netherlands, Switzerland, United Kingdom)

This playbook is the "how" that goes with the Atlas data. Every program-specific fact below comes from the Atlas, where each value links to an exact quote on the program's official page. General advice is labelled as advice. Nothing here is immigration, visa or legal advice. For those topics, use only official government sources (see Section 13).

1. Start here (10 minutes)

1. Open **Atlas.html** in any browser. It works offline. Filter by what blocks you most (Physics GRE, English-test waiver, fee waivers), then star programs to build a shortlist.
2. Open **Atlas.xlsx** (Excel, Google Sheets, or LibreOffice). Pick your starred programs in the **Shortlist** tab. Deadlines, fees and rules fill in automatically. The **Budget**, **Recommenders** and **Supervisor outreach** tabs do the rest of the tracking.
3. Read Sections 2 to 7 of this playbook today. They cover the decisions with deadlines attached: fee waivers, English tests, GRE scores, Canadian supervisors, and Europe's different system.
4. Before you pay any fee or book any test, click the value in the Atlas and open the official page. Programs change rules mid-cycle. When the Atlas says *Not stated*, the pages we checked did not say. It does **not** mean "no".

2. Your remaining timeline (late-October start)

Treat these as working targets (advice). The fixed dates come from the programs.

When	Do this
This week	Shortlist 8 to 15 programs. Check which need the **Physics GRE** (Section 5) and whether your English-test plan covers every program (Section 4).
By **Nov 1, 2026**	If Rice is on your list, request its worldwide **Doctoral Application Fee Waiver Weeks** waiver; Rice says it then needs a completed application by **Nov 27**.

When	Do this
Oct 31 / Nov 1	EPFL's autumn round closes Oct 31 (next round Apr 30). Heidelberg's physics graduate-school portal closes **Nov 1**.
Early November	Email Canadian supervisors (Section 6), and the European professors or advertised positions you want (Section 7). Send recommenders their packet (Section 10).
Nov 01: Heidelberg (HGSFP); Nov 30: Ohio State; Dec 01: Alberta; Dec 01: UBC	The earliest deadlines in this Atlas. Several programs review applications the moment the deadline passes.
Mid-December	Most US deadlines fall on Dec 11-16 (see the deadline table). Cambridge's funding deadline is **Dec 8**.
January	Most Canadian deadlines (Jan 7-15), plus UIUC, Penn State, Texas A&M and Rutgers. Edinburgh's physics funding deadline is Jan 22.
Feb-April	Interviews and offers. Several programs in this Atlas (e.g., Stony Brook and Texas A&M) ask admitted students to respond by **April 15**. SISSA's spring call closes in early March; EPFL's second round on Apr 30.

Deadlines in this Atlas (earliest first)

Deadline	Program	Country	Note
Nov 01, 2026	Heidelberg (HGSFP)	Germany	year inferred; act by 2026-11-01
Nov 30, 2026	Ohio State	USA	year inferred
Dec 01, 2026	Alberta	Canada	year inferred
Dec 01, 2026	UBC	Canada	
Dec 01, 2026	UT Austin	USA	year inferred
Dec 03, 2026	Columbia	USA	
Dec 08, 2026	Cambridge	United Kingdom	
Dec 11, 2026	Maryland	USA	
Dec 11, 2026	Toronto	Canada	year inferred
Dec 14, 2026	Stanford	USA	
Dec 14, 2026	UChicago	USA	
Dec 15, 2026	Boston University	USA	year inferred
Dec 15, 2026	Caltech	USA	year inferred
Dec 15, 2026	Cornell	USA	year inferred
Dec 15, 2026	Georgia Tech	USA	year inferred
Dec 15, 2026	McGill	Canada	year inferred
Dec 15, 2026	Purdue	USA	year inferred
Dec 15, 2026	Rice	USA	

Deadline	Program	Country	Note
Dec 15, 2026	Stony Brook	USA	year inferred
Dec 15, 2026	Berkeley	USA	
Dec 15, 2026	Minnesota	USA	year inferred
Dec 15, 2026	UW-Madison	USA	
Dec 16, 2026	UCSB	USA	
Jan 01, 2027	Rutgers	USA	year inferred
Jan 07, 2027	Queen's	Canada	year inferred; rolling
Jan 09, 2027	Waterloo	Canada	year inferred
Jan 10, 2027	Texas A&M	USA	
Jan 15, 2027	Penn State	USA	year inferred
Jan 15, 2027	SFU	Canada	year inferred; rolling
Jan 15, 2027	Calgary	Canada	year inferred
Jan 15, 2027	UIUC	USA	
Jan 22, 2027	Edinburgh	United Kingdom	
Feb 01, 2027	McMaster	Canada	year inferred; rolling
Apr 30, 2027	EPFL	Switzerland	year inferred
Not stated	Munich (MCQST)	Germany	
Not stated	SISSA	Italy	
Not stated	Ottawa	Canada	
Per position	ETH Zurich	Switzerland	
Per position	Leiden	Netherlands	
Rolling	Bonn-Cologne (BCGS)	Germany	

"Year inferred" means the page gave a day and month but no year; we assumed the Fall 2027 cycle. Confirm on the page.

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3. Build a balanced list, and a budget

Advice. Aim for 8 to 15 applications. Choose by research fit first and prestige second. A program where two or more faculty work on what you want is a stronger application than a famous name where nobody does.

- **Fit test:** for each program, name two faculty members and one recent paper from each that you could discuss. If you can't, it isn't a fit yet.

- **Spread:** mix programs where your profile is typical of admits with a few reaches. Use each department's own descriptions (several state typical GPAs or class sizes on their pages) rather than forum rumours.
- **Cost reality:** the median international application fee in this Atlas is **95 USD** for US programs and **150 CAD** for Canadian programs (where stated), so twelve applications cost roughly twelve times that before test-score fees. Budget it in the **Budget** tab, and add score-sending and transcript costs from the official sites. We do not estimate those for you.

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4. English tests in 2026: what changed and who waives

Verified change: TOEFL iBT moved to a new **1-6 score scale on January 21, 2026**. Programs now list two minimums, one per scale (e.g., UBC: 90 on the old scale or 4.5 on the new one). Check which scale your score is on.

There are four kinds of waiver rule. Which kind a program uses decides whether an English-medium Indian degree is enough:

1. **Friendly to an English-medium Indian degree ("Yes" in the Atlas):** EPFL, McGill, Rutgers, Stanford, Stony Brook, UBC, UT Austin, Toronto. Most of these look at your university's language of instruction. Some ask for an official letter or transcript notation (e.g., Toronto), and some for documents sent to the department (McGill). UT Austin is different: its exempt-country list includes India itself.
2. **Conditional / list-based.** Exemption only if your institution meets a specific test. Alberta publishes a list that includes the IITs (all campuses), IISc and the University of Hyderabad but **not** the IISERs. Waterloo accepts only institutions listed on WHED with English as the *only* language of instruction, and explicitly rejects medium-of-instruction letters. Queen's lets you *request* an exemption after a recent year of English-medium study. In this Atlas: Columbia, McMaster, Queen's, Alberta, Edinburgh, Waterloo.
3. **No waiver for an English-medium Indian degree ("No"):** Georgia Tech, SFU, Ohio State, Calgary, Berkeley, Cambridge, UIUC. These either use an "English-primary country" list that India is not on (e.g., UIUC, Berkeley, SFU, Calgary, and Cambridge via the UK government's list), or require a test from everyone (Georgia Tech).
4. **Not stated** on the pages checked: ask the program before you assume.

Advice: if even one program on your list needs a score, book the test now. Score delivery takes time; Waterloo, for example, says it takes 4-6 weeks for official scores to reach it. Also check speaking minimums. Several programs tie teaching-assistant eligibility (and so part of your funding) to a separate, higher speaking score.

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5. GRE and Physics GRE: required again at some top programs

Most programs in this Atlas don't need the GRE, and several will **not even look** at it. A notable group requires the **Physics GRE** for this cycle:

- **Caltech**: Physics GRE REQUIRED for the Fall 2027 cohort

- **Cornell**: REQUIRED for theory applicants (theoretical astro, particle, condensed matter); theory applications without it cannot be considered
- **Stanford**: Physics GRE REQUIRED for all applicants
- **Calgary**: REQUIRED for all foreign applicants
- **Berkeley**: Physics GRE REQUIRED for Fall 2027 entry
- **UChicago**: Physics GRE REQUIRED for the 2026-2027 cycle

Programs that will not consider **any** GRE score, even if you send one: Rice, Stony Brook, UCSB, Minnesota. Programs that ignore the **General** GRE but still review the Physics GRE: UT Austin, UW-Madison.

Advice:

- If your list contains any program in the first group, you need a Physics GRE score that arrives **before that program's deadline**. Check the remaining ETS test dates and score-reporting times on ets.org today; don't rely on forum dates.
- If you have a strong score and a program marks it *Optional* or *Recommended*, sending it can help. UT Austin is a special case: it asks you to explain in your Statement of Purpose why your Physics GRE matters, or it ignores the score.
- If you have no score and your list has none of the "Required" programs, don't spend scarce weeks on it now.

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6. Canada works differently: supervisors and master's-first rules

Supervisor first

In Canada, admission often depends on a professor agreeing to take you. In this Atlas:

Program	Supervisor before admission	What the page says
McGill	Recommended	Not required, but highly recommended: professors select students for their groups
McMaster	Recommended	Funding includes a research fellowship from the supervisor's grant; contact faculty directly to ask if they are taking students
Queen's	Recommended	A complete application needs a research proposal; work with a potential supervisor early
SFU	Not stated	Not stated in the text checked; transfers need supervisory committee approval
Alberta	Not stated	Physics pages do not state a pre-application supervisor requirement; FGPS notes some thesis programs require one
UBC	Not required	Name faculty of interest in the application; no prior commitment needed, but contacting faculty is encouraged
Calgary	Required	No supervisor needed to apply, but you are admitted only if at least one potential supervisor will fund you; contacting faculty first helps
Ottawa	Required	Supervisor's name required AT APPLICATION; you must be accepted by a thesis supervisor to register

Program	Supervisor before admission	What the page says
Toronto	Not required	No supervisor needed to apply; MSc and 5-year direct-entry PhD students pick one after arrival; 4-year PhD entrants (with a master's) must secure one before registration, typically by May
Waterloo	Required	Admission requires a faculty member willing to supervise you; contact before or after applying and name supervisors on the application

Master's-first rules

Many Canadian PhD programs expect a master's degree. A BSc holder usually starts in the MSc and may transfer to the PhD later. In this Atlas, "Master's usually required": McGill, Queen's, SFU, UBC, Ottawa, Waterloo. Direct or either route: Alberta, Toronto.

Important for 5-year integrated BS-MS students (IISER, IISc, IIT and others): whether your degree counts as a master's equivalent is decided by each university's credential assessment. **Ask each program in writing.** Don't assume either way. Put the question in your first email to the graduate office and keep the reply.

Funding is stated more often

Canadian programs often publish guaranteed minimums (e.g., UBC at least CAD 40,000 a year for years 1-4; Alberta CAD 100,000 over four years; McMaster at least CAD 21,000 after tuition and fees). Note whether a figure is before or after tuition.

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7. Europe works differently

The 9 European programs in this Atlas (Germany, Italy, Netherlands, Switzerland, United Kingdom) use two very different ways in.

1. **Graduate-school calls with deadlines.** You apply to a school or program by a published date: Heidelberg's HGSFP (portal closes Nov 1), Cambridge (funding deadline Dec 8), Edinburgh (funding deadline Jan 22), SISSA (exam-based spring call, early March), Munich's MCQST fellowships (call opens October 2026), and EPFL (rounds close Oct 31 and Apr 30, but you also need a thesis advisor to offer you a position).
2. **Apply to a professor or an advertised paid position.** There is no central committee. At ETH Zurich each professor recruits their own students. Leiden fills paid positions advertised as vacancies. The Bonn-Cologne graduate school says it does not provide PhD positions: you find one yourself. In this Atlas, position- or supervisor-based: EPFL, ETH Zurich, Leiden, Bonn-Cologne (BCGS).

Program	Country	Deadline	How you get in	Funding
EPFL	Switzerland	Apr 30, 2027	Admission = eligible by committee AND a thesis advisor offers you a PhD position after interview; eligible files are shared with all thesis directors	Enrolment requires secured funding; lab-appointed doctoral students get an employment contract

Program	Country	Deadline	How you get in	Funding
ETH Zurich	Switzerland	Per position	A professor must agree to supervise you; contact professorships directly (the Doctoral Administration does not match candidates)	With few exceptions doctorates are linked to a position (employment) at ETH or in the ETH Domain
Heidelberg (HGSFP)	Germany	Nov 01, 2026	Two routes: the web portal (international students, or anyone without funding) or direct application with an advisor and funding already arranged	The portal route is for applicants without prior financial support; funding terms are not stated on the pages checked
Munich (MCQST)	Germany	Not stated	Not stated on the page checked	Doctoral fellowships; terms not stated on the page checked
Leiden	Netherlands	Per position	Apply to a specific advertised position; requirements are set with the thesis supervisor	All LION PhD projects are 4-year fully paid positions (usually a 1+3-year contract) with about 10% teaching
SISSA	Italy	Not stated	Selection is a competition (qualifications, written exam, interview); no prior supervisor requirement mentioned	Not stated on the page checked
Bonn-Cologne (BCGS)	Germany	Rolling	BCGS does not provide PhD positions: find and apply for a PhD position (with a supervisor) at Bonn or Cologne yourself	Funding comes with the PhD position you secure; not stated centrally
Cambridge	United Kingdom	Dec 08, 2026	Some PhD programmes need supervisor contact before applying; a 2027-28 supervisor list is published	Not guaranteed: apply by the funding deadline to enter the University funding competition; otherwise secure your own support
Edinburgh	United Kingdom	Jan 22, 2027	No need to contact supervisors before applying unless you propose your own project; name preferred supervisors on the form	School studentships include full tuition; UKRI limits how many international students can be funded, but other routes exist

Fees are often low or zero. EPFL charges no application fee. Cambridge charges £20 per applicant per year for up to 7 doctoral applications.

A master's is usually expected first. In this Atlas: EPFL, Heidelberg (HGSFP), Leiden, Cambridge. There are exceptions:

- Heidelberg has an integrated 4+4 master/doctoral track for excellent candidates with four years of physics.
- EPFL accepts 4- or 5-year bachelor's holders who explain their education in the Statement of Objectives.
- Edinburgh says most of its current PhD students had no master's before starting.

If you hold a 5-year BS-MS, ask each program how it assesses your degree, and keep the reply.

Funding often means a job contract. Leiden's PhD projects are 4-year paid positions. At EPFL, enrolment requires secured funding and lab-appointed students get an employment contract. ETH doctorates are, with few exceptions, linked to a position. In the UK, funding is competitive: Cambridge runs a funding competition with a December deadline, and Edinburgh notes limits on how many international students UKRI grants can fund. In this Atlas, paid or typically funded: EPFL, ETH Zurich, Leiden; competitive: Cambridge, Edinburgh.

English rules in the UK are strict for Indian degrees. Cambridge's exemption depends on the UK government's list of majority English-speaking countries, and India is not on it. Edinburgh accepts English-taught degrees only from institutions on its approved list. EPFL requires English proficiency but makes no test mandatory (it strongly recommends TOEFL).

Advice.

- In position-based systems, your supervisor email (Section 8) *is* the application. Check the vacancy pages weekly.
- Allow time after an offer for contracts and permits. EPFL says about 3 months is usually needed for visas and work permits.

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8. Supervisor emails that get answered (advice)

Find people, not just programs. Use faculty pages, the group's own website (look for "accepting students"), and papers from the last two years. Shortlist 2-4 people per university.

The 6-sentence email:

> **Subject:** Prospective PhD student (Fall 2027): [specific topic] in your group
>
> 1. Who you are: degree, institution, expected graduation.
> 2. The specific work of theirs you read, and one genuine question or connection.
> 3. Your most relevant research experience and what *you* did in it.
> 4. One skill or result that would help their group (a technique, code, measurement).
> 5. The ask: "Are you planning to take new PhD students for Fall 2027, and would you be open to my application?"
> 6. Thanks, with CV and transcript attached as PDFs named Lastname_CV.pdf.

Cadence: one polite follow-up after about 10 days if you get no reply, then move on. Log every email in the **Supervisor outreach** tab; it calculates the follow-up date for you.

Don'ts: no mass emails with swapped names, no long life stories, no attachments over a few MB, and no generic praise ("your esteemed research").

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9. Statement of purpose: a physics framework (advice)

Programs set their own length rules. Alberta, for example, wants **500 words or less on a single page**. Read each program's prompt and obey it exactly.

A structure that works for research PhDs:

1. **Opening (2-3 sentences):** the physics question that drives you, stated concretely ("how disorder changes spin dynamics in..."), not "I have loved physics since childhood".
2. **Research experience (40-50% of the text):** for each project, give the question, *your* contribution, the method, the result, and what you learned, including what went wrong and how you fixed it.
3. **Skills and preparation:** techniques, instruments, code, and advanced courses that map onto their work.
4. **Why this program:** two named faculty and why, plus facilities or centres that matter for your plans.
5. **Goals:** where the PhD leads and why this training fits.

Check before you submit: every sentence should be true and specific, and you should be able to defend it in an interview. Write it yourself. Never use purchased or ghost-written statements.

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10. Recommendation letters: the request kit (advice)

- **Who:** research supervisors first, then instructors of advanced courses who know your work. Programs differ on number: Toronto wants **exactly two** (no more), while UIUC, Columbia and Maryland list three.
- **When:** ask at least 4-6 weeks before your earliest deadline, and send portal invitations early.
- **What to send (one email, one PDF):** your shortlist with deadlines and portal links, CV, draft SOP, a "brag sheet" (projects, dates, what you did, skills shown), and a reminder of specific moments they saw your work.
- **Reminders:** a friendly note 10 days and again 3 days before each deadline. The **Recommenders** tab tracks status per program.

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11. Fee waivers, ethically

What this Atlas found for international applicants:

- **Free or waived for everyone this cycle:** Penn State. **No application fee at all:** EPFL.
- **Open to internationals (need, hardship, or a time-boxed window):** Columbia, Cornell, Rice.
- **Limited or conditional routes:** Caltech, Georgia Tech, Purdue, Rutgers, Stanford, Ohio State, UBC, Cambridge, UChicago, UT Austin.
- **Domestic applicants only:** Boston University, Stony Brook, Texas A&M, Berkeley, UCSB, UW-Madison.
- **No waivers:** Alberta, Calgary, Toronto.

Advice:

- Request a waiver **before** paying. Application fees are typically non-refundable (UChicago's Physical Sciences Division says so for its $90 fee), so a fee paid too early is gone.
- Ask briefly and honestly: state your circumstances and the program you're applying to, and attach evidence if asked. Never misstate citizenship, income or event attendance.
- Some departments have small waiver budgets that run out months before the deadline (Rutgers says so), so ask early.

12. Submission checklist

☐ Program-specific statement prompt answered (length, format, required sections)
☐ CV in PDF, with research section, publications/posters, skills
☐ Transcripts uploaded for every post-secondary institution (some programs, e.g., Toronto, ask for official copies only after admission)
☐ English score sent from the testing agency, or waiver evidence ready
☐ GRE / Physics GRE sent if required (check institution codes on each page)
☐ Recommenders invited, and each has confirmed they will submit
☐ Fee paid or waiver approved, with the receipt saved
☐ Supervisor names listed where the program asks (Waterloo, Ottawa, UBC and others)
☐ Final check of every value in the Atlas against the live official page

13. Money, safety and what this kit does not cover

- **Scams:** be wary of anyone who guarantees admission, sells "special" fee-waiver codes, or asks for your portal passwords. Real waiver codes come from the program itself; Penn State, for example, publishes its own code on its admissions page.
- **Paying fees:** many portals take Visa or Mastercard credit cards. Some accept debit cards selected as "credit" (UBC says this). Enable international online transactions with your bank before the deadline week.
- **Visas and study permits:** out of scope. Use only official sources: travel.state.gov for US student visas, canada.ca (IRCC) for Canadian study permits, and for Europe the host university's international office and the national immigration authority it points you to. Paid immigration advice in Canada is a regulated activity, and this kit is not that.

14. How this data was built (and how to report an error)

- Each of the 40 programs was checked against its own department and graduate-school pages on October 21, 2026. Every value stores the exact sentence it came from, the URL and the date. A script confirms that each quote appears verbatim on the fetched page.
- Research and drafting were AI-assisted. Every published value is tied to an official quote, and uncertain items are labelled *Not stated*, *Conditional*, *year inferred* or *sources conflict* instead of guessed.
- A weekly automated check re-reads the cited pages and flags quotes that disappear. Updates are sent to buyers as new versions at no charge.

- A few well-known European programs (for example, Oxford and ISTA Austria) block automated checks, so they are not in this edition. They will be added once their pages can be verified.
- **Found something wrong or changed?** Reply to your purchase email with the program and the page link. Confirmed corrections ship in the next update.